

Evolutionary Trajectory: From Physical to Molecular Regulation

Protocell

Turgor pressure
Simple geometry
Basic chemistry

LUCA

Physical constraints
Basic FtsZ
Supercoiling

Modern Bacteria

Sophisticated
regulation
Redundancy
Checkpoints



Evolution adds molecular complexity on top of physical foundations

Regulation shifts from: PURELY PHYSICAL → PHYSICAL + BASIC MOLECULAR → SOPHISTICATED MOLECULAR (on physical foundation)